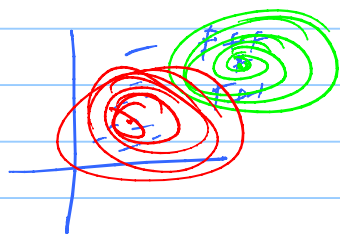


COL774 / COL734
(ML) Aug 30, 2025

UDA :-

Gaussian Discriminant Analysis

$$0 \leq \phi \leq 1$$



$$p(x^{(i)}, y^{(i)}; \theta)$$

$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$$y \sim \text{Bernoulli}(\phi)$$

$$x|y=1 \sim N(\mu_1, \Sigma_1)$$

$$= \frac{1}{(2\pi)^{d/2} |\Sigma_1|^{1/2}} e^{-\frac{(x-\mu_1)^T \Sigma_1^{-1} (x-\mu_1)}{2}}$$

$\nwarrow$
 $x|y=0$

$$x|y \sim N(\mu_y, \Sigma_y)$$

$$y \in \{0, 1\}$$

$$\theta = (\phi, \mu_0, \Sigma_0, \mu_1, \Sigma_1) \quad (1)$$

$$x|y=1$$

$$x|y=0 \sim N(\mu_0, \Sigma_0)$$

$$x|y=1 \sim N(\mu_1, \Sigma_1)$$

Log. likelihood?

(Generative Models) $P(x^{(i)}, y^{(i)}; \theta)$

as opposed to $P(y^{(i)} | x^{(i)}; \theta)$

for logistic (Discriminative Models)

$$\mathcal{L}(\theta) = \log \prod_{i=1}^m [P(x^{(i)}, y^{(i)}; \theta)]$$

using i.i.d. assumption over set of data points



$$P(y^{(i)}; \theta) \quad P(x^{(i)} | y^{(i)}; \theta)$$

$$\begin{aligned} \mathcal{L}(\theta) = & \sum_{i=1}^m \left[1 \{y^{(i)} = 1\} \log(\phi) \right] + \sum_{i=1}^m \left[1 \{y^{(i)} = 0\} \log(1 - \phi) \right] \\ & + \sum_{i=1}^m \left[1 \{y^{(i)} = 1\} \log P(x^{(i)} | y^{(i)} = 1; \theta) \right] \end{aligned}$$

(1) (2) (3)

$$+ \sum_{i=1}^m [1\{y_i = 0\} \log p(x_i | y_i = 0; \mu_0, \Sigma_0)] \quad (9)$$

H.W. $\Rightarrow$ for ① $\nabla_{\phi} \mathcal{L}(\theta) = 0$ easy

② $\nabla_{\mu_0}, \nabla_{\mu_1}$ of $\mathcal{L}(\theta) = 0$ (Moderate) / easy

Analytically
can make
gradient 0

③ $\nabla_{\Sigma_0}, \nabla_{\Sigma_1}$ of $\mathcal{L}(\theta) = 0$

$$\phi = \frac{\sum_{i=1}^m 1\{y_i = 1\}}{\sum_{i=1}^m 1\{y_i = 1\} + \sum_{i=1}^m 1\{y_i = 0\}}$$

$$\frac{\phi}{1-\phi} = \frac{\sum_{i=1}^m 1\{y_i = 1\}}{\sum_{i=1}^m 1\{y_i = 0\}}$$

$$\mu_1 = \frac{\sum_{i=1}^m 1\{y_i = 1\} x_i}{\sum_{i=1}^m 1\{y_i = 1\}}$$

similarly for μ_0

$$\hat{\Sigma}_1 = \frac{\sum_{i=2}^m \sum_{j=2}^m y_{ij}^w = 13 \quad \underbrace{(x_{i1}^w - \mu_1)(x_{j1}^w - \mu_1)^T}_{\sum_{i=2}^m \sum_{j=2}^m y_{ij}^w = 13}}{\sum_{i=2}^m \sum_{j=2}^m y_{ij}^w = 13} \quad \downarrow$$

$$\hookrightarrow (\hat{\Sigma}_1)_{jk} =$$

$$\text{Emp. cov}(X_1, X_k) = \frac{\sum_{i=2}^m \sum_{j=2}^m y_{ij}^w (x_{i1}^w - \mu_1)(x_{jk}^w - \mu_{1k})}{\sum_{i=2}^m \sum_{j=2}^m y_{ij}^w = 13}$$

Similarly for $\hat{\Sigma}_0$

Suppose $\hat{\Sigma}_0 = \hat{\Sigma}_1 = \hat{\Sigma}$ (Assumption)

$$\hat{\Sigma} = \frac{\sum_{i=2}^m \underbrace{(x_{i1}^w - \mu_{y_{i1}})(x_{i1}^w - \mu_{y_{i1}})^T}_m}{m}$$

↳ Inference (Test Time)
 x

$$\underline{\theta}^* = (\phi^*, \mu_1^*, \Sigma_1^*, \mu_0^*, \Sigma_0^*)$$

$$p(y=1 | x; \theta)$$

$$\Rightarrow \log \left[\frac{p(y=1 | x; \theta)}{p(y=0 | x; \theta)} \right] = 0 \quad (\text{Decision Boundary})$$

① In general LHS is quadratic in x
 $\Rightarrow$ Quadratic decision boundary

② If $\Sigma_0 = \Sigma_1 = \Sigma$, then LHS is linear in x
 $\Rightarrow$ Linear decision boundary

③ $\rightarrow p(y=1|x; \theta)$ takes the form of
 logistic function.
 when $(\Sigma_0 = \Sigma_1 = \Sigma)$

$$\rightarrow \text{LHS} = \log \left[\frac{\frac{1}{(2\pi)^{d/2}} |\Sigma_1|^{1/2} e^{-\frac{(x-x_1)^T \Sigma_1^{-1} (x-x_1)}{2}}}{\frac{1}{(2\pi)^{d/2}} |\Sigma_0|^{1/2} e^{-\frac{(x-x_0)^T \Sigma_0^{-1} (x-x_0)}{2}}} \cdot \left(\frac{\phi}{1-\phi} \right) \right] \quad \text{--- ①}$$

$$\rightarrow \log \frac{p(y=1|x; \theta)}{p(y=0|x; \theta)} = \log \left[\frac{p(x|y=1; \theta) p(y=1; \theta)}{p(x|y=0; \theta) p(y=0; \theta)} \right]$$

$$p(y|x; \theta) = \frac{p(x, y; \theta) p(y; \theta)}{p(x; \theta)}$$

$$\left[\frac{p(x; \theta)}{p(x; \theta)} \right]$$